CANDIDATE NAME (CG) ________________________ *INDEX NUMBER* _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2011

BIOLOGY
Higher 1

Paper 2

Friday
12 August 2011

2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST
Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

.
For Section A, write your answers in spaces provided on the question paper.
For Sections B, write your answers in the writing paper provided.

At the end of examination,
1. fasten all your work securely together;

FOR EXAMINER'S USE	
Section A	
1	**/9**
2	**/10**
3	**/11**
4	**/7**
5	**/3**
Section B	
5/6	**/20**
TOTAL	**/60**

INFORMATION FOR CANDIDATES
The intended number of marks is given in brackets [ ] at the end of each question or part question.

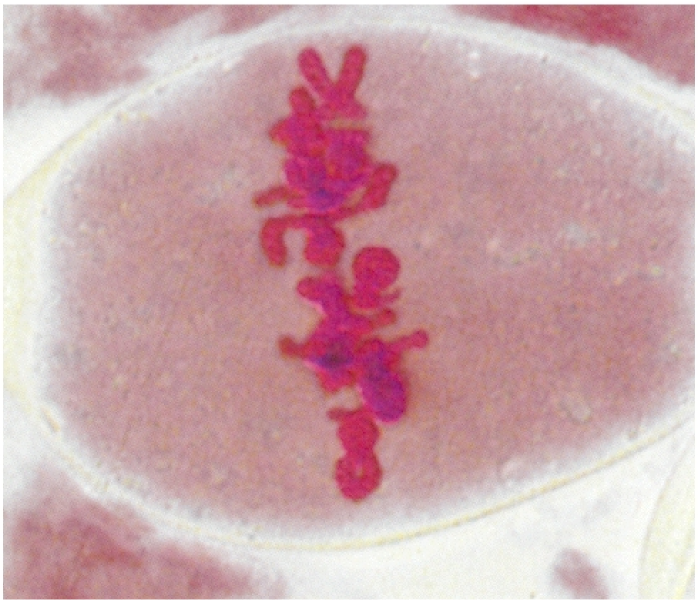
This question paper consists of 12 printed pages

Section A (40 marks)

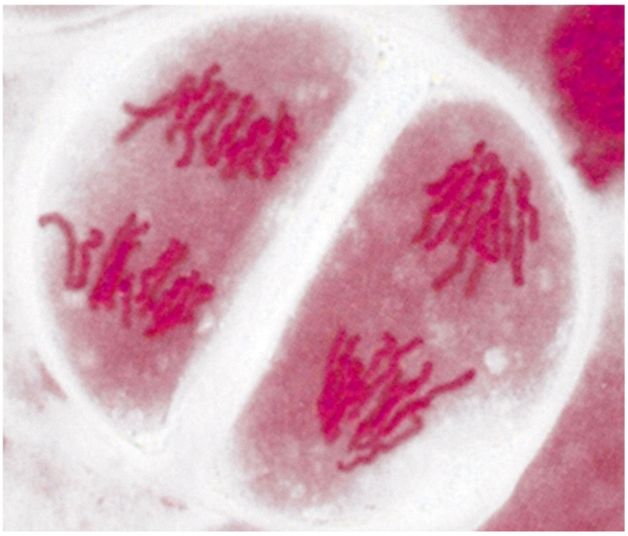
Answer all questions in the space provided

Question 1

Fig. 1.1 shows a cell undergoing two different stages of meiosis (to the same scale).



A



B

Fig. 1.1

(a) Identify stages **A** and **B**. [1]

Stage A ……………………………………….. Stage B ………………………………….

(b) Explain the phenomenon observed in **B**. [2]

……………………………………………………………………………………………………

……………………………………………………………………………………………………

(c) Describe how meiosis can lead to genetic variation. [2]

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

(d) Suggest a possible outcome of a disorganised metaphase plate. [1]

……………………………………………………………………………………………………

……………………………………………………………………………………………………

The chromosome carrying alleles **F/f** also carries alleles **E/e**, as shown in **Fig 1.2**.

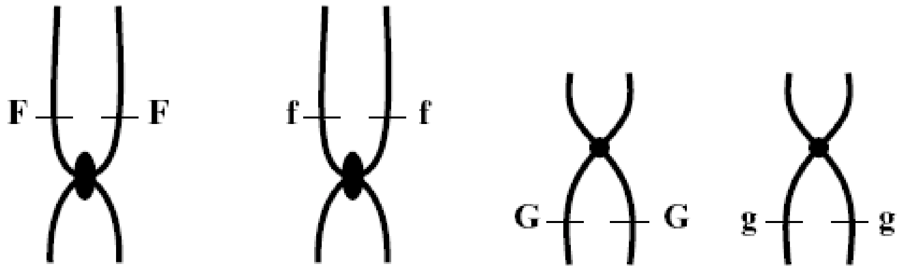


Fig. 1.2

(e)(i) State the genotype of this cell. [1]

……………………………………………………………………………………………………

(ii) Explain how there are two copies of each allele of each gene at this stage of nuclear division. [2]

……

……

[Total: 9 marks]

Question 2

Fig. 2.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

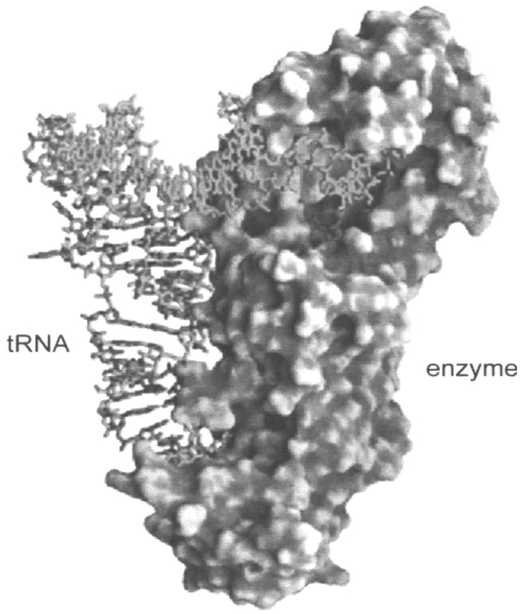


Fig. 2.1

(a) Describe **two** structural features which adapt tRNA to its role in translation. [2]

……

……

……

……

(b) Explain how the enzyme in **Fig. 2.1** is suited to its function. [3]

………………………………………………………………………………………………………

………………………………………………………………………………………………………

………………………………………………………………………………………………………

………………………………………………………………………………………………………

(c) **Fig. 2.2** below shows the base sequence on a length of mRNA. The key indicates which amino acids they represent.

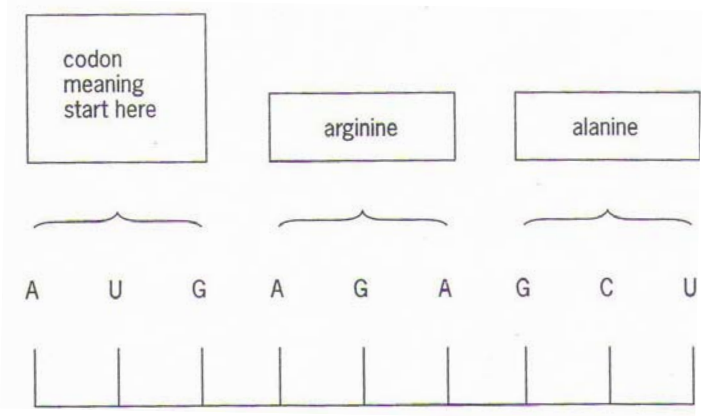


Fig. 2.2

The amino acid arginine is shown here as AGA but can also be represented by the codon AGG.

(i) Explain why the substitution of A by G in this case does not affect the amino acid represented. [1]

………………………………………………………………………………………………………

………………………………………………………………………………………………………

(ii) The codons in the mRNA sequence do not overlap. Suggest 2 advantages of this. [2]

………………………………………………………………………………………………………

………………………………………………………………………………………………………

Some genetic drugs are short sequences of nucleotides. They act by binding to selected sites on DNA or mRNA molecules and preventing the synthesis of disease-related proteins. There are two types:

1. triplex drugs are made from DNA nucleotides and bind to the DNA forming a three stranded helix.
2. antisense drugs are made from RNA nucleotides and bind to mRNA.

(iii) Name the process in protein synthesis which will be inhibited by [2]:

Drug type	**Process inhibited**
Triplex drugs	
Antisense drugs	

[Total: 10 marks]

Question 3

Fig. 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

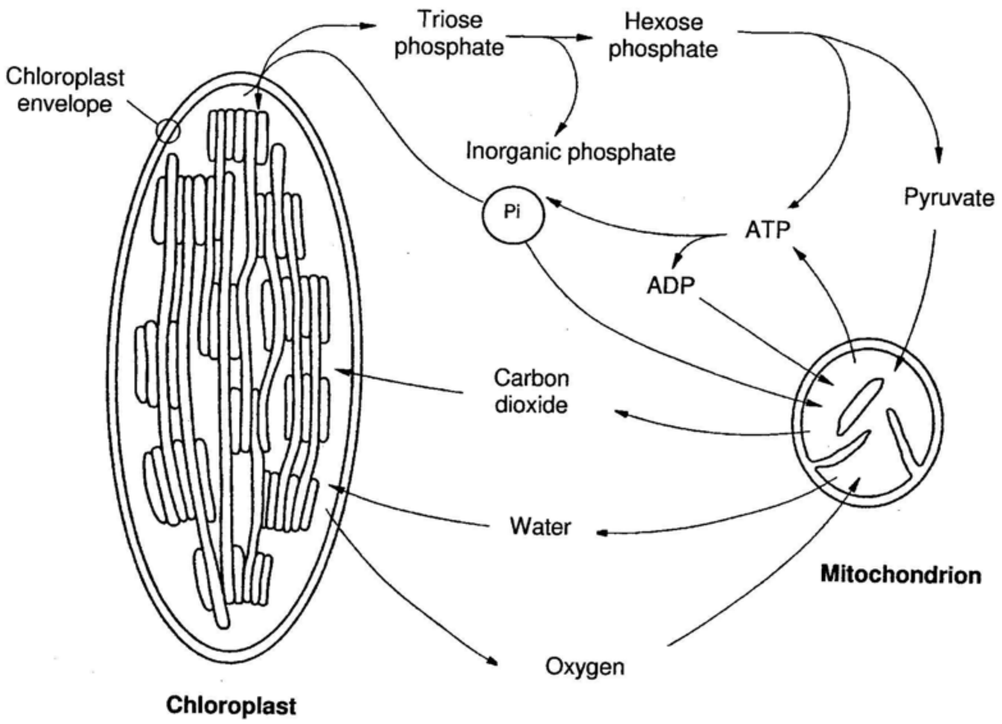


Fig. 3.1

A leafy shoot is sealed inside an air-tigh … t container. The concentration of oxygen in the atmosphere within this contai … e measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 3.1**, outline how the following links the metabolic pathways between the chloroplast and mitochondrion of this cell.

(i) triose phosphate and hexose phosphate; [2]

……

……

……

(ii) gaseous exchange. [2]

……

……

……

(b) With reference to **Fig. 3.1**,

(i) explain how it is possible for the oxygen concentration to remain constant. [1]

……

……

(ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism? [2]

……

……

……

In plants, the carbohydrates manufactured are channeled to form two types of macromolecules, namely cellulose and starch.

(c) State the two structural differences between cellulose and starch. [2]

……

……

……

(d) Explain how these differences allow them to serve different functions in a plant cell. [2]

……

……

……

[Total: 11 marks]

Question 4

In 1905, the Dutch botanist Hugo De Vries, in an experiment to test Mendel's rediscovered work, found that when a red-flowered, three-leaved clover plant was crossed with a white-flowered, five-leaved plant the following offspring were produced:

36 red-flowered, five leaved
41 red-flowered, three-leaved
40 white-flowered, five-leaved
38 white-flowered, three-leaved

The alleles for red flowers and five leaves are dominant.

(a) Using the symbols **R** for red flowers, **r** for white flowers, **E** for five leaves and **e** for three leaves, draw a genetic diagram to explain these results. [5]

(b) When these plants were self-pollinated, explain why only the white flowered, three-leaved plants bred true. [2]

………

………

………

[Total: 7 marks]

Question 5

A new brightly-coloured mouse was discovered in Russia and it was given the name *Mus colores*. **Fig. 3.1** below shows two flowcharts that illustrate how blue and green cells were isolated from different parts of this mouse, and each cell was cultured individually.

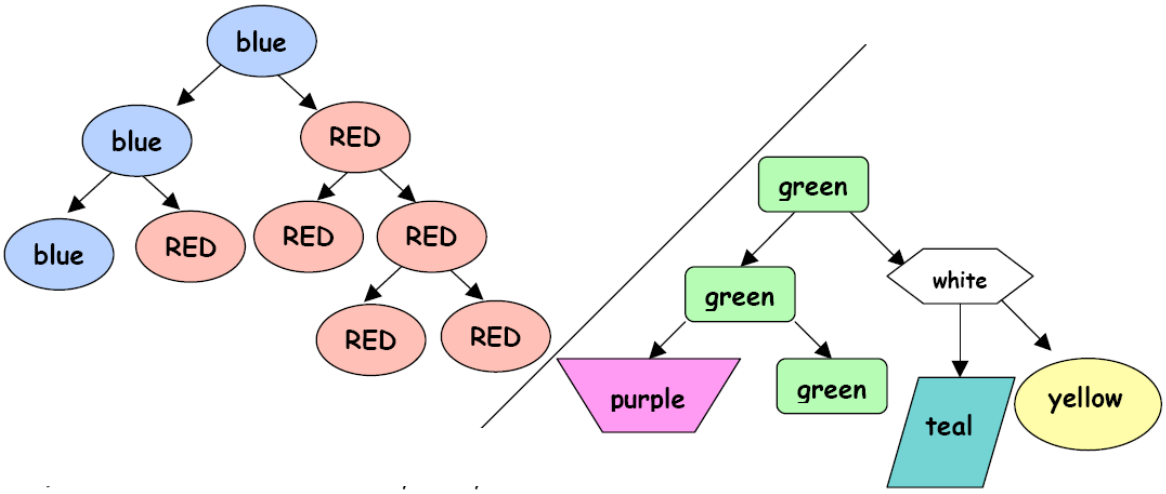


Fig. 3.1

(a) With reference to **Fig. 3.1**,

(i) are the blue cells stem cells? Give a reason for your answer. [1]

……………………………………………………………………………………………

……………………………………………………………………………………………

(ii) given that the green cell is a stem cell, explain why green cells can give rise to differently coloured cells. [2]

……………………………………………………………………………………………

……………………………………………………………………………………………

……………………………………………………………………………………………

Section B (20 marks)

Answer **one** question.
Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 5

(a) Explain, with examples, how genetic engineering can improve the quality and yield of crop plants and animal. [14]

(b) Describe the properties of plasmids that would allow them to be use as vectors. [6]

Question 6

(a) Describe and explain the secondary structure of proteins. [6]

(b) Describe the main ways in which a globular protein differs from DNA. [6]

(c) Compare and contrast between DNA replication and transcription. [8]